Normal Parent-Hamiltonian Range Reduction in Cirac–Perez-Garcia–Schuch–Verstraete 2021

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1 The assertion

The range-reduction step for normal matrix product states appears in Section IV.C of Cirac–Pérez-García–Schuch–Verstraete [CPGSV21].[1]

Let A be a normal MPS tensor, and let L_0 be its injectivity length: after blocking L_0 consecutive sites, the resulting tensor is injective. The source first proves an intersection step for two overlapping local Hamiltonian terms of range L_0+1. These terms act on sites $1,\dots,L_0+1$ and $2,\dots,L_0+2$. The source denotes the three regions by A, B, and C: the first site, the middle region consisting of sites $2,\dots,L_0+1$, and the last site. The tensor associated to the middle region B is injective. For a state in the intersection

$$\mathcal{G}_{1,\dots,L_0+1} \cap \mathcal{G}_{2,\dots,L_0+2},$$

the argument inverts the left boundary tensor A_{left}, restores the middle region B, and then uses injectivity of the enlarged region to invert the joint region. It concludes that every vector in the intersection already lies in $\mathcal{G}_{1,\dots,L_0+2}$; the reverse inclusion is immediate. [2]

The source then says that this argument can be iterated. Hence the ground space obtained from all length-L_0+1 terms agrees with the ground space obtained from all longer length-L_0+k terms. When $k=L_0$, the proof either appeals to the previous $2L_0$-range uniqueness theorem or applies the same kind of argument when closing the periodic boundary. The resulting theorem states that a normal MPS which becomes injective after blocking L_0 sites has a unique ground state for the parent Hamiltonian whose interaction range is L_0+1. [3]

[1] For traceability, this note corresponds to issue #1042. The principal formal obligation is recorded in issue #588; the wrapped-boundary comparison is recorded in issue #961; and the parent-Hamiltonian consequences are recorded in issue #460.

[2] Source location: `Papers/2011.12127/TN-Review-main.tex`:2049–2077, especially the middle-region argument around the displayed figures following `eq:4:initiate-intersection-proof`.

[3] Source location: `Papers/2011.12127/TN-Review-main.tex`:2078–2079 for the iteration and boundary-closing sentence, and `Papers/2011.12127/TN-Review-main.tex`:2087–2090 for the theorem labeled `thm:4:unique-gs-L0_plus_1`.

2 The formal statement

The corresponding formal statement keeps the intended hypotheses. In mathematical terms, it assumes that A is normal, that A is injective after blocking L_0 sites, and that the periodic chain has length $N \geq 2$ with an interaction range L satisfying $L_0 < L \leq N$. The desired conclusion is that the periodic-chain ground space is the line spanned by the periodic MPS vector $\Omega_N(A)$:

$$\mathcal{G}_{N,L}(A) = \mathbb{C}\,\Omega_N(A).$$

In the formal statement this is the equality between the cyclic chain ground space and the MPV, or periodic MPS vector, subspace. The hard inclusion into the MPV line is the remaining range-reduction lemma. [4]

The blueprint records the same range-reduction target in `blueprint/src/chapter/ch14_parent_hamiltonian.tex`:1212–1240, under the label `thm:normal_range_reduction_containment`. Its proof sketch already separates the open-chain containment from the cyclic boundary comparison, which is the same division made by the formal statements below.

The first mismatch is that the available intersection theorem is not a literal expression of the proof through the region B. That theorem was written for a tensor that is already injective at one site. Its proof decomposes the identity as a linear combination of the one-site matrices A_j and then iterates this single-letter intersection property through contiguous windows. [5] For a normal tensor with injectivity length L_0, the available inverse is the inverse for words of length L_0, not a one-site inverse. This is not a contradiction in the source argument; it is a mismatch between the diagrammatic region-by-region proof and the available formal statements for intersection properties.

The formal argument therefore separates the proof into two parts instead of reproducing the source paragraph verbatim. The open-chain component has been proved by a block-injective range-reduction lemma: if every contiguous window of length $L_0 + 1$ lies in the corresponding local MPS ground space, then the full open chain lies in the full MPS ground space. Combining this with cyclic window monotonicity gives an open-chain consequence from the cyclic length-L constraints. This supplies the formal analog of the iterated open-boundary part of the source proof, without expressing the inverse for the region B through the older single-site intersection lemma. [6]

[4] Formal locations: `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:651–660 for `MPSTensor.chainGroundSpace_le_mpvSubmodule_of_normal_range_reduction`, and `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:670–680 for `MPSTensor.chainGroundSpace_eq_mpvSubmodule_normal`. The formal hypotheses are named `IsNormalA` and `IsNBlkInjectiveAL0`, together with `2<=N` and `L0<L<=N`.

[5] Formal locations: `TNLean/MPS/ParentHamiltonian/IntersectionProperty.lean`:253–256 for `MPSTensor.groundSpace_intersection`, `TNLean/MPS/ParentHamiltonian/IntersectionProperty.lean`:313–330 for the use of `decompositionMaphA1`, and `TNLean/MPS/ParentHamiltonian/CyclicWindow.lean`:153–155 for the contiguous iteration using that theorem.

[6] Formal locations: `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:347–353 for `MPSTensor.contiguous_mem_groundSpace_of_isNBlkInjective`, and `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:409–424 for `MPSTensor.chainGroundSpace_le_groundSpace_of_isNBlkInjective`.

3 The remaining comparison

After the open-chain step, a periodic-chain ground state can be written with an open-chain boundary matrix X. The point left to prove is that the wrapped length-$L_0 + 1$ constraints force X to be scalar, and hence that the state is in the MPV line. The formal proof has extracted the two one-sided wrapped compatibilities. In schematic notation, the wrapped window gives

$$C_\tau^+ A_j X = Y_\tau^+ A_j,$$

while the mirror wrapped window gives

$$X A_j C_\tau^- = A_j Y_\tau^-.$$

Here τ denotes the complementary boundary word exposed by a wrapped window, equivalently the physical labels outside the open middle interval. The formal wrapped and mirrored lemmas produce analogous complement-word products, written schematically as C_τ^+ and C_τ^-, but the two products are obtained with different orientations and indexings. The two formulas are both valid; what is missing is a comparison that puts the two outputs over one common middle word and one common witness family. [7]

The algebraic endgame needs a stronger same-witness statement. There should be a common middle word μ and a single family of matrices Y_μ such that, for every physical letter j,

$$A^\mu A_j X = Y_\mu A_j, \qquad X A_j A^\mu = A_j Y_\mu.$$

Once these two identities have the same middle word and the same witness, the formal algebraic lemma proves that X commutes with all words of a fixed positive length; block injectivity then promotes this to commutation with the full matrix algebra, so X is scalar. [8] The remaining point is the comparison between the two wrapped-window outputs: one must reindex their complementary words to a common middle word and identify the two witnesses as a single same-witness family.[9]

4 Conclusion

The comparison is provisional. It identifies a difference between the compressed source paragraph and the available formal lemmas; it is not a source-error claim.

[7]Formal locations: `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:522–534 for `MPSTensor.chainGroundSpace_wrapped_boundary_compatibilities_of_isNBlkInjective`, `TNLean/MPS/ParentHamiltonian/WrappingWindow.lean`:386–397 for the wrapped compatibility, and `TNLean/MPS/ParentHamiltonian/WrappingWindow.lean`:434–445 for the mirror compatibility.

[8]Formal locations: `TNLean/MPS/ParentHamiltonian/WrappingWindow.lean`:527–535 for `MPSTensor.commutes_words_of_two_sided_middle_compatibility`, and `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean`:629–643 for `MPSTensor.groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective_of_two_sided_middle_compatibility`.

[9]For traceability, this is the mathematical content recorded in issue #961.

The cited Cirac–Perez-Garcia–Schuch–Verstraete 2021 argument is not refuted. The source compresses two mathematical steps which the formal proof must keep separate: the B-region invert-and-grow open-chain range reduction, and the boundary-closing comparison of the witnesses arising from the wrapped windows.

The open-chain part has been proved under the intended block-injective hypotheses, though not by reusing the older single-site intersection theorem. The last mathematical point is the same-witness comparison at the periodic boundary. If that comparison is proved, the difference is only expository. If it reveals an additional mathematical obstruction, the conclusion should be revised accordingly.

References

[CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.